



7169 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Mosaics	Regular			

JWST Proposal 7169 (Created: Wednesday, July 16, 2025, 2:00:24PM Eastern Standard Time) - Overview

Folder	Observation	Label	Observing Template	Science Target
	1	Arches Cluster - Regular - Band 2 - Reduced Leakcals	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	2	Arches Cluster - Regular - Band 2 - Reduced Leakcals	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	3	Arches Cluster - Regular - Band 2 - Reduced Leakcals	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	4	Arches Cluster - Regular - Band 2 - Reduced Leakcals	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	5	Arches Cluster - Regular - Band 2	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	6	Arches Cluster - Regular - Band 2	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	7	Arches Cluster - Regular - Band 2	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	8	Arches Cluster - Regular two-in-one - Band 3 - Reduced Leakcal	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	9	Arches Cluster - Regular two-in-one - Band 3 - Reduced Leakcal	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	10	Arches Cluster - Regular two-in-one - Band 3 - Reduced Leakcal	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	11	Arches Cluster - Regular two-in-one - Band 3 - Reduced Leakcal	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	12	Arches Cluster - Regular two-in-one - Band 3	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	13	Arches Cluster - Regular two-in-one - Band 3	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	14	Arches Cluster - Regular two-in-one - Band 3	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	15	Arches Cluster - Regular two-in-one - Band 3	NIRSpec IFU Spectroscopy	(1) CenterIFUArches

Folder	Observation	Label	Observing Template	Science Target
	16	Arches Cluster - Regular two-in-one - Band 3	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
Mosaics - Bright Objects				
	17	Arches Cluster - Bright Objects two-in-one - Band 3 - Reduced Leakage	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	18	Arches Cluster - Bright Objects two-in-one - Band 3	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	19	Arches Cluster - Bright Objects two-in-one - Band 3	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	20	Arches Cluster - Bright Objects two-in-one - Band 3	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	21	Arches Cluster - Bright Objects - Band 2	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	22	Arches Cluster - Bright Objects - Band 2	NIRSpec IFU Spectroscopy	(1) CenterIFUArches
	23	Arches Cluster - Bright Objects - Band 2 - Reduced Leakage	NIRSpec IFU Spectroscopy	(1) CenterIFUArches

ABSTRACT

Located at the centre of the Milky Way, the Arches cluster is a metal-rich, young massive star cluster, pervaded by massive stars that populate the cluster mass-function up to the highest masses. Its proximity, compactness, and youth, combined with the unprecedented spectroscopic sensitivity of JWST NIRSpec, enables --for the first time-- a detailed census and characterization of its heavily reddened massive cohort. We propose NIRSpec IFU spectroscopic observations covering 1.66 to 5.27 μm of a significant fraction of the Arches, down to early B-dwarfs, at a modest observational cost. Our results will allow for the first comprehensive test of the validity of current models of radiation driven winds in a high-metallicity environment. We will also obtain a direct measurement of the extinction towards each star, breaking current degeneracies that plague ground-based and space photometric studies. Quantitative spectroscopic analysis of the rich suite of H-, He-, and metal-line diagnostics with state-of-the-art model atmosphere programs will return accurate fundamental stellar properties, including masses and mass-loss rates. This will provide a gauge of the mechanical energy, momentum and ionizing power provided by the cluster to its environment, as well as the cluster mass function and its potential top-heavy nature. Consequently, the Arches cluster will become the prototypical template for understanding clusters and their environments in more

distant, metal-rich, massive star-forming galaxies.

OBSERVING DESCRIPTION

The program consists of a NIRSpec IFU mosaic with 29 tiles of 3.0"x3.0". The mosaic is composed of two different kinds of tiles.

The provided coordinates roughly match the cluster center (e.g. Hosek et al. 2022).

The proper motion of the cluster is small, of the order of 1 mas/yr (Hosek et al. 2022), hence negligible for the observational window.

-Tiles with bright objects:

Grating: G235H

Blocking Filter: F170LP

Aperture: 0.11"

Dithering pattern: 4-Point-Dither

Detector strategy: FULL-frame with NRSRAPID read-out pattern. 5 groups per integration. 4 integrations per exposure.

Grating: G395H

Blocking Filter: F290LP

Aperture: 0.11"

Dithering pattern: 4-Point-Dither

Detector strategy: FULL-frame with NRSRAPID read-out pattern. 2 groups per integration. 8 integrations per exposure.

-Regular tiles:

Grating: G235H

Blocking Filter: F170LP

Aperture: 0.11"

Dithering pattern: 4-Point-Dither

Detector strategy: FULL-frame with NRSIRS2RAPID read-out pattern. 14 groups per integration. 1 integrations per exposure.

JWST Proposal 7169 (Created: Wednesday, July 16, 2025, 2:00:24PM Eastern Standard Time) - Overview

Grating: G395H

Blocking Filter: F290LP

Aperture: 0.11"

Dithering pattern: 4-Point-Dither

Detector strategy: FULL-frame with NRSIRS2RAPID read-out pattern. 9 groups per integration. 1 integrations per exposure.

The mosaic has been designed for a specific V3PA = 90 degrees angle.

No TA is requested. The pointing accuracy delivered by the observatory is enough.

Our observational window coincides with the Micrometeroid Avoidance Zone (MAZ).

Proposal 7169 - Targets - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)	Proper Motion RA: -0.96 mas/yr	
			Dec: -28 49 19.93 (-28.82220d)	Proper Motion Dec: -2.29 mas/yr	
			Equinox: J2000	Epoch of Position: 2000	
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible. Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES				

Proposal 7169 - Observation 1 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 1: Arches Cluster - Regular - Band 2 - Reduced Leakcals											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.46											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)				Proper Motion RA: -0.96 mas/yr					
			Dec: -28 49 19.93 (-28.82220d)				Proper Motion Dec: -2.29 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
Template	Description=[O clusters, Open star clusters, Young star clusters]											
	Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	14	2	false	true	NONE	4	8	1750.667	256619.53
	2	G235H/F170LP	NRSIRS2RAPID	14	2	true	false	NONE	1	2	437.667	256619.53

Proposal 7169 - Observation 1 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 2 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 2: Arches Cluster - Regular - Band 2 - Reduced Leakcals											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.46											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)				Proper Motion RA: -0.96 mas/yr					
			Dec: -28 49 19.93 (-28.82220d)				Proper Motion Dec: -2.29 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	14	2	false	true	NONE	4	8	1750.667	256619.53
	2	G235H/F170LP	NRSIRS2RAPID	14	2	true	false	NONE	1	2	437.667	256619.53

Proposal 7169 - Observation 2 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 3 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 3: Arches Cluster - Regular - Band 2 - Reduced Leakcals											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSspec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.46											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	14	2	false	true	NONE	4	8	1750.667	256619.53
	2	G235H/F170LP	NRSIRS2RAPID	14	2	true	false	NONE	1	2	437.667	256619.53

Proposal 7169 - Observation 3 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 4 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 4: Arches Cluster - Regular - Band 2 - Reduced Leakcals											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSspec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.46											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	14	2	false	true	NONE	4	8	1750.667	256619.53
	2	G235H/F170LP	NRSIRS2RAPID	14	2	true	false	NONE	1	2	437.667	256619.53

Proposal 7169 - Observation 4 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 5 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 5: Arches Cluster - Regular - Band 2											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.46											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	14	2	false	true	NONE	4	8	1750.667	256619.53
	2	G235H/F170LP	NRSIRS2RAPID	14	2	true	true	NONE	4	8	1750.667	256619.53

Proposal 7169 - Observation 5 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 6 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 6: Arches Cluster - Regular - Band 2											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.46											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)				Proper Motion RA: -0.96 mas/yr					
			Dec: -28 49 19.93 (-28.82220d)				Proper Motion Dec: -2.29 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
Template	Description=[O clusters, Open star clusters, Young star clusters]											
	Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	14	2	false	true	NONE	4	8	1750.667	256619.53
	2	G235H/F170LP	NRSIRS2RAPID	14	2	true	true	NONE	4	8	1750.667	256619.53

Proposal 7169 - Observation 6 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 7 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 7: Arches Cluster - Regular - Band 2											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.46											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2RAPID	14	2	false	true	NONE	4	8	1750.667	256619.53
	2	G235H/F170LP	NRSIRS2RAPID	14	2	true	true	NONE	4	8	1750.667	256619.53

Proposal 7169 - Observation 7 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 8 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 8: Arches Cluster - Regular two-in-one - Band 3 - Reduced Leakcal											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.44											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000			Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible. Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES		
	Template	TA Method						HFF Readout Mode				
NONE						false						
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	256619.48
	2	G395H/F290LP	NRSIRS2RAPID	9	2	true	false	NONE	1	2	291.778	256619.48

Proposal 7169 - Observation 8 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 9 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 9: Arches Cluster - Regular two-in-one - Band 3 - Reduced Leakcal											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpect IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.44											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000			Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible. Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES		
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	256619.48
	2	G395H/F290LP	NRSIRS2RAPID	9	2	true	false	NONE	1	2	291.778	256619.48

Proposal 7169 - Observation 9 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 10 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 10: Arches Cluster - Regular two-in-one - Band 3 - Reduced Leakcal											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSspec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.44											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	256619.48
	2	G395H/F290LP	NRSIRS2RAPID	9	2	true	false	NONE	1	2	291.778	256619.48

Proposal 7169 - Observation 10 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 11 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 11: Arches Cluster - Regular two-in-one - Band 3 - Reduced Leakcal											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.44											
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)				Proper Motion RA: -0.96 mas/yr					
			Dec: -28 49 19.93 (-28.82220d)				Proper Motion Dec: -2.29 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
Template	Description=[O clusters, Open star clusters, Young star clusters]											
	Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	256619.48
	2	G395H/F290LP	NRSIRS2RAPID	9	2	true	false	NONE	1	2	291.778	256619.48

Proposal 7169 - Observation 11 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 12 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 12: Arches Cluster - Regular two-in-one - Band 3											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpect IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.44											
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	256619.48
	2	G395H/F290LP	NRSIRS2RAPID	9	2	true	true	NONE	4	8	1167.111	256619.48

Proposal 7169 - Observation 12 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 13 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 13: Arches Cluster - Regular two-in-one - Band 3											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpect IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.44											
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)				Proper Motion RA: -0.96 mas/yr					
			Dec: -28 49 19.93 (-28.82220d)				Proper Motion Dec: -2.29 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
Template	Description=[O clusters, Open star clusters, Young star clusters]											
	Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	256619.48
	2	G395H/F290LP	NRSIRS2RAPID	9	2	true	true	NONE	4	8	1167.111	256619.48

Proposal 7169 - Observation 13 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 14 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 14: Arches Cluster - Regular two-in-one - Band 3											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.44											
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	256619.48
	2	G395H/F290LP	NRSIRS2RAPID	9	2	true	true	NONE	4	8	1167.111	256619.48

Proposal 7169 - Observation 14 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 15 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 15: Arches Cluster - Regular two-in-one - Band 3											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.44											
Diagnostics	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)				Proper Motion RA: -0.96 mas/yr			Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible. Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES		
	Dec: -28 49 19.93 (-28.82220d)				Proper Motion Dec: -2.29 mas/yr							
	Equinox: J2000				Epoch of Position: 2000							
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	256619.48
	2	G395H/F290LP	NRSIRS2RAPID	9	2	true	true	NONE	4	8	1167.111	256619.48

Proposal 7169 - Observation 15 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 16 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 16: Arches Cluster - Regular two-in-one - Band 3											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpect IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.44											
Diagnostics	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000			Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible. Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES		
	Template	TA Method						HFF Readout Mode				
NONE						false						
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	9	2	false	true	NONE	4	8	1167.111	256619.48
	2	G395H/F290LP	NRSIRS2RAPID	9	2	true	true	NONE	4	8	1167.111	256619.48

Proposal 7169 - Observation 16 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 17 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 17: Arches Cluster - Bright Objects two-in-one - Band 3 - Reduced Leakcal											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.49											
Diagnostics	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)				Proper Motion RA: -0.96 mas/yr			Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible. Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES		
			Dec: -28 49 19.93 (-28.82220d)				Proper Motion Dec: -2.29 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	4	8	false	true	NONE	4	32	1717.883	256619.51
	2	G395H/F290LP	NRSRAPID	4	8	true	false	NONE	1	8	429.471	256619.51

Proposal 7169 - Observation 17 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 18 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 18: Arches Cluster - Bright Objects two-in-one - Band 3											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.49											
Diagnostics	(Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)				Proper Motion RA: -0.96 mas/yr					
			Dec: -28 49 19.93 (-28.82220d)				Proper Motion Dec: -2.29 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible. Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	4	8	false	true	NONE	4	32	1717.883	256619.51
	2	G395H/F290LP	NRSRAPID	4	8	true	true	NONE	4	32	1717.883	256619.51

Proposal 7169 - Observation 18 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 19 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 19: Arches Cluster - Bright Objects two-in-one - Band 3											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.49											
Diagnostics	(Visit 19:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	4	8	false	true	NONE	4	32	1717.883	256619.51
	2	G395H/F290LP	NRSRAPID	4	8	true	true	NONE	4	32	1717.883	256619.51

Proposal 7169 - Observation 19 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 20 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 20: Arches Cluster - Bright Objects two-in-one - Band 3											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.49											
Diagnostics	(Visit 20:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		50.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSRAPID	4	8	false	true	NONE	4	32	1717.883	256619.51
	2	G395H/F290LP	NRSRAPID	4	8	true	true	NONE	4	32	1717.883	256619.51

Proposal 7169 - Observation 20 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 21 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 21: Arches Cluster - Bright Objects - Band 2											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.52											
Diagnostics	(Visit 21:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000			Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible. Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES		
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSRAPID	10	4	false	true	NONE	4	16	1889.672	256619.47
	2	G235H/F170LP	NRSRAPID	10	4	true	true	NONE	4	16	1889.672	256619.47

Proposal 7169 - Observation 21 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 22 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 22: Arches Cluster - Bright Objects - Band 2											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.52											
Diagnostics	(Visit 22:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d) Dec: -28 49 19.93 (-28.82220d) Equinox: J2000				Proper Motion RA: -0.96 mas/yr Proper Motion Dec: -2.29 mas/yr Epoch of Position: 2000					
	Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible.											
	Category=Stellar Cluster											
	Description=[O clusters, Open star clusters, Young star clusters] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSRAPID	10	4	false	true	NONE	4	16	1889.672	256619.47
	2	G235H/F170LP	NRSRAPID	10	4	true	true	NONE	4	16	1889.672	256619.47

Proposal 7169 - Observation 22 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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Proposal 7169 - Observation 23 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Observation	Proposal 7169, Observation 23: Arches Cluster - Bright Objects - Band 2 - Reduced Leakcals											Wed Jul 16 19:00:24 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: Additional ETC was done to verify the SNR of the fainter targets: 256619.52											
Diagnostics	(Visit 23:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CenterIFUArches	RA: 17 45 50.6783 (266.4611596d)				Proper Motion RA: -0.96 mas/yr			Comments: The proper motion (Hosek et al. 2022) could offset the mosaic and the targets by 0.0001 mas in RA and 0.0003 mas in DEC the mosaic and the targets. This is negligible. Category=Stellar Cluster Description=[O clusters, Open star clusters, Young star clusters] Extended=YES		
	Dec: -28 49 19.93 (-28.82220d)				Proper Motion Dec: -2.29 mas/yr							
	Equinox: J2000				Epoch of Position: 2000							
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	11	11	0.0		0.0		0.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSRAPID	10	4	false	true	NONE	4	16	1889.672	256619.47
	2	G235H/F170LP	NRSRAPID	10	4	true	false	NONE	1	4	472.418	256619.47

Proposal 7169 - Observation 23 - Dissecting the Arches cluster: An ideal test bed to study the metal-rich Universe

Special Requirements	Aperture PA Range 228.97164917 to 228.97164917 Degrees (V3 90.0 to 90.0)
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